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REGULAR ARTICLE

The global stakeholder capitalism model of digital platforms and its implications for strategy and innovation from a Schumpeterian perspective

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Abstract
In this article, we introduce the global stakeholder capitalism model of digital platforms. We describe the role that big data plays in the formation of the new extended innovation ecosystems of digital platforms and explain how these extended innovation ecosystems give rise to this new model. We discuss the implications of this new model for the strategy of digital platforms from a Schumpeterian perspective and show how digital platforms profit from innovation by capturing a form of rents that differs from standard Schumpeterian rents rooted in innovation economics. We conclude that the sustainability of this model depends on a new form of governance that assumes multi-fiduciary corporate duties in the extended innovation ecosystems of digital platforms.

Keywords Strategy · Innovation · Digital platforms · Innovation ecosystems

JEL classification: O30 · O31 · O32 · O33 · O36

1 Introduction

Advances in machine learning are among the key drivers behind the rise of digital platforms and the emergence of commercial-grade applications enabled by Artificial Intelligence and offered by digital platforms (Brynjolfsson and McAffee 2017). The curation of user-generated content, the generation of big data through the facilitation of interactions among the different sides served by digital platforms, and the implementation of a new class of efficient deep learning algorithms fueled by big data and

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run on highly scalable parallel computing architectures are key forces behind the so-called digital platform revolution. A growing body of research in business economics has also unveiled the potential to generate and harness the power of demand-side economies of scale as a crucial factor that drives the new business models of digital platforms (Spulber 2019; van Alstyne et al. 2006).

Besides understanding better the technological drivers of digital platforms and the economic underpinnings of multi-sided markets, we also need to understand the new model under which digital platforms operate from a perspective that goes beyond technology and markets in order to unveil not only the current opportunities and threats but also the future development of digital platforms.

Digital platforms operate globally at a supranational level. To unveil the new model under which they operate, we thus need to analyze digital platforms not from a purely technological or economic perspective. We rather need to use a more general construct at the intersection of corporate strategy, business ethics, politics and international relations. Following such an integrative approach, we set out in this article the model under which digital platforms operate today, explaining not only the implications of this model for the strategy and innovation of digital platforms from a neo-Schumpeterian perspective but also its implications along the political and international relations dimensions.

We describe how the generation of large volumes of big data as a core asset leads to the extension of the innovation ecosystems of digital platforms. We also explain why the positioning of big data as a generalized complementary asset in these innovation ecosystems allows digital platforms to deal with the deep strategic uncertainty and lack of organizational alignment that pervades the exploration of new uses for increasingly large volumes of big data. We also compare the strategy of digital platforms under this new model with mainstream strategic management frameworks rooted in the resource-based view of the firm (Barney 1991; Penrose 1995) and on the positioning framework (Porter 2008) and describe how the way digital platforms profit from innovation departs from the way in which firms profit from innovation under the standard model (Teece 1986, 2006, 2018a, b). Finally, we explain the role that a new form of extended governance of digital platforms will play in the future and why such extended governance will replace sustainable competitive advantage as the essence of digital platform strategy.

The article is organized as follows. In Section 2, we briefly introduce key concepts of digital platforms, revisit the notion of core and complementary assets from the perspective of digital platforms and then discuss the Teecean model of how firms profit from innovation. In Section 3, we present a number of cases of incumbent digital platforms, analyzing how they profit from innovation by extending their innovation ecosystems. Based on the cases analyzed in Section 3, we introduce in Section 4 the model under which digital platforms operate today, which we refer to as the "global stakeholder capitalism model of digital platforms." In Section 5, we discuss the main implications of our model for the strategy and innovation of digital platforms from a Schumpeterian perspective and describe how they profit from innovation in a way that departs from the standard model (Teece 1986) and its subsequent revisions (Teece 2006, 2018a, b). In Section 6, we present our conclusions and ideas for future work.

2 Background

We begin this section by introducing digital platforms and the concepts of core and complementary assets as they relate to the innovation strategy of digital platforms.

2.1 Digital platforms

Although there is no agreed-upon definition of what a digital platform is, several key characteristics of digital platforms have been proposed. Parker et al. (2016) define digital platforms in terms of properties such as: "that platforms benefit from network effects, especially network effects that generate demand-side economies of scale, that they allow the launching of complementary offerings, that they allow third-parties to transact through the platform, or that they drive innovations through open innovation."

The term platform is widely used today to refer to different ways of organizing value creation through Internet technologies in hypercompetitive and knowledge-intensive industries (D'Aveni 1994, 2010; D'Aveni et al. 2010). Other scholars distinguish between internal and external platforms, noting that internal platforms are predicated on "modular architectures that deploy reusable components" and where value creation takes place primarily within the firm, even when relying on an extensive network of suppliers (Gawer and Cusumano 2014, p. 419).

But these definitions offer neither a satisfactory explanation of what digital platforms are nor a description of how they function. In fact, many business models that predated the Internet share many of the characteristics of what we refer to as digital platforms. A more satisfactory definition can be given if we analyze business models in networked economies from an evolutionary perspective.

The business models that predated the Internet and operated in networked economies shared one common characteristic, namely, they based their competitive advantage in providing the best network infrastructure. These business models relied on owning and controlling the access to network infrastructure as a core and bottleneck asset, were very capital intensive, and deployed broadcasting services based on one-to-many connectivity in a world where content was not yet plentiful. The value created by these business models grew linearly with the nodes (users) in the network according to what became known as the law of Sarnoff.[1] The competitive advantage in these business models was based on content delivered through the best proprietary network infrastructure and was measured in terms of network coverage rather than the amount of content delivered.

With the advent of the Internet in the mid 1990s this situation changed dramatically. As a global network infrastructure available to Internet entrepreneurs and their competitors and also to users for free, the source of competitive advantage shifted away from the best network infrastructure to the provision of the best content measured in terms of quality and quantity. As a network of networks, the Internet allowed

[1] Named after David Sarnoff, a pioneer of broadcasting networks in the U.S.

for the first time to interconnect users, making the economic value of the network grow based on stronger network effects. Colloquially referred to as the Web 1.0, this first generation of Internet business models drove their competitive advantage based on the maxim that "content is king" and generated network effects governed by what became known as the law of Metcalfe (Hendler and Golbeck 2008).[2]

Many of today's digital platforms started as Web 1.0 applications, including well-known platforms such as Amazon, which initially owned inventory of books to then sell them on the Internet. For all intent and purposes, Amazon was not a platform then but rather an e-retailer that operated with the logic of Web 1.0 applications because it owned and controlled access to content.[3] Amazon thus operated with the logic of linear value chains of the Web 1.0, where the value is created by the firm and the full potential of users and other third-parties to create value was not yet unleashed.

The transition of Amazon from its original business model to the new model that ignited its process of hyper growth took place once it had reached a critical mass of clients. Once this milestone was achieved, Amazon started to intermediate transactions in the long tail (Anderson 2006) using a business model that carried over the legacy of value creation in a linear value chain. While it continued to keep inventory in the short tail, Amazon started to rely on providers instead of users as a source of content based on strong competencies in input and output logistics for order intake and order fulfillment. The same applied to Netflix, which relied on third-party content not contributed by users as a source of competitive advantage. But while Netflix had the advantage of providing digital media over the Internet, which eliminated the complex problem of input and output logistics of hard goods, it had the disadvantage of having to negotiate bespoke contracts with the firms owning copyrighted content.

The key to the success of these Internet business models was the variety of third-party content, comprised of hard goods, digital media or a combination of both, which were made available to clients/users over the Internet. This enabled a new form of pull marketing models based upon offering a large variety of items catering to clients/users in the long tail, which allowed a form of auto segmentation of users/clients not yet seen in the business models that predated the Internet (Benghozi and Benhamou 2010; Hinz et al. 2011). As these new Internet business models construe value creation as taking place in linear value chains, some scholars refer to them as supply-chain platforms (Gawer 2014). But the central feature of digital platforms is not the control of value creation in linear value chains.

Rather than controlling existing linear value chains, digital platforms create value creation systems and orchestrate value creation processes in them (Normann 2001). In these value creation systems, value is created not only by the platform but also by third parties that are part of these value creation systems and are usually referred to as the "sides" of the platform. Although these value creation systems are based on open

[2] According to this law, the economic value of the network could grow to the square of the number of nodes (users) if all users in the network are equally valuable in terms of enabling interconnections (Briscoe et al. 2006).

[3] Which in the case of the initial business model of Amazon corresponded to books.

innovation, open innovation systems were already a key component of the business models based on linear value chains that predated the Internet (Chesbrough 2003). We will use the term innovation ecosystems to refer to the value creation systems of digital platforms as opposed to the open innovation systems of linear value chains.

The process of value creation in the innovation ecosystems of digital platforms is usually associated with the concept of user-generated content in group-forming networks. The content contributed by the sides of the platform in these group networks is not only limited to digital content but can also encompass hard goods as well. Colloquially referred to as the Web 2.0, group-forming network applications are governed by the law of the pack, which is also referred to as the law of Reed.[4] But the Web 2.0 is a more general and ambitious project that aims to reposition the entire Internet as an innovation platform (O'Reilly 2007).

Some scholars distinguish different types of digital platforms based on two contrasting views, namely, the economics and the engineering views. As noted by Gawer (2014, p. 1239), while the economics view construes digital platforms as "multi-sided markets and focuses on competition," the engineering view construes them as "technological architectures with a focus on innovation." Gawer classifies platforms in three organizational settings, namely, internal platforms within a firm, supply-chain platforms operating in linear value chains, and industry platforms operating in ecosystems (Gawer 2014, p. 1244). The classification of platforms in these categories aims to provide a "unified conceptualization" of platforms based on the view that "the different types of platforms described in the different platform literatures ... constitute in fact various manifestations of an underlying similar phenomenon" (Gawer 2014, p. 1245). We argue, however, that industry platforms operating in complex innovation ecosystems constitute quite a departure from the other two categories and that their strategy and innovation departs in fundamental ways from the strategy and innovation of internal and supply-chain platforms.

Without making a strict separation between the economics and engineering view mentioned above, we focus in this article on multi-sided digital platforms operating in complex innovation ecosystems construed as value creation systems as opposed to linear value chains. Accordingly, we define digital platforms as "platforms that drive their competitive advantage by offering the best facilitation of transactions and interactions to multiple sides through the platform so that they can create and diffuse value in value creation systems through open innovation."

Our aim in this article is not only to introduce the model under which digital platforms, as we have defined them above, operate but also to describe how this model may evolve as we enter a new era in the development of the Internet. In order to understand why this new era in the evolution of the Internet will be based on the extension of the current innovation ecosystems of digital platforms beyond the sides they serve today, we need to introduce the new role that big data will play as a core and also as a complementary asset in what we will refer to as the "extended innovation ecosystems" of digital platforms.

[4] According to which the economic value generated by these applications grows exponentially with the number of nodes (users) in the network (Reed 2001).

2.2 Big data as a core and complementary asset of digital platforms

Based on the subsymbolic paradigm of Artificial Intelligence, big data allows the development of commercial-grade AI-enabled solutions in application domains that are not yet knowledge intensive and can thus be successfully addressed using data-driven as opposed to knowledge-based approaches (Devlin 1999). It is through big data in the form of hundreds of millions of (user) models that digital platforms are in the privileged position to deploy data analytics, machine learning, user modeling and natural language processing technologies for a reason that is of the utmost importance for the sustainability of their business models, namely, to dramatically increase the density of offerings they can deliver to “audiences of one” (Normann 2001). These technologies allow digital platforms to unlock and harness the power of the big data harvested by digital platforms on a global scale.

Our analysis strongly suggests that big data, as the high-octane fuel needed to render Artificial Intelligence technologies useful, is the core asset that enables digital platforms to increase the density of offerings in their innovation ecosystems. It also suggests that Artificial Intelligence technologies can be positioned as a core or as a complementary asset in the innovation ecosystems of digital platforms to profit from innovation.

This analysis begs a question of high strategic importance, namely, whether big data should be kept captive as the core asset of digital platforms or whether it should (also) be positioned as a complementary asset that digital platforms make available in their innovation ecosystems so that third parties can become drivers of innovation and profit from it. The latter positioning may sound controversial, as it runs counter to the standard model of how firms profit from innovation (Teece 1986, 2006, 2018a, b).

The innovation ecosystems orchestrated by digital platforms allow them to drive open innovation by bringing new sides on board as complementors. But digital platforms can also engage other interested parties in the provision of services that lie outside the scope of their original business. The provision of these services by interested parties takes place in what we will refer to as the extended innovation ecosystems of digital platforms, which are comprised of heterogenous groups of interested parties to which dominant digital platforms provide big data as a generalized complementary asset.

The role of generalized complementary assets in profiting from innovation was discussed by Teece (1986). In his article, he set out strategies that firms should pursue to access complementary assets and profit from innovation. In the remainder of this section, we briefly discuss his model.

2.3 The standard model of how firms profit from innovation

In its original article describing how firms profit from innovation (Teece 1986), and its subsequent revisions (Teece 2006, 2018a, b), David Teece summarized the entry strategies of innovative firms under strong and weak appropriability regimes. The more common case that is of interest from the perspective of digital platforms is the one in which the dominant design of an innovation has not yet emerged under a weak appropriability regime (Teece 2006, p. 1139). In such situations, and in case digital platforms were not able to promote a dominant design internally, the model proposed by Teece suggests that digital platforms should wait until the dominant design emerges during the preparadigmatic phase. As will be shown in Section 3 and discussed in Section 5, this would generally be a strategic folly on the part of digital platforms that they should avoid by creating the conditions for a dominant design to emerge in their innovation ecosystems without delay (Arthur 2012).

The logic of the Teecean model stipulates further that, in case digital platforms were able to promote a dominant design and if the complementary assets needed to profit from a given innovation were internally available, then digital platforms should commercialize the innovation without delay. The more interesting cases for digital platforms, however, are those in which such complementary assets are not available internally. In such cases, the next relevant question in the Teecean model is whether such complementary assets are specialized or not.[5]

In case the needed complementary assets were not specialized, the Teecean model would encourage digital platforms to subcontract for the development of such assets. This assumes that there is an efficient market where these specialized assets are freely available. However, such cases are rather exceptional because these assets need to be cospecialized and are therefore not generally available. Moreover, developing these cospecialized assets is ridden with deep strategic uncertainty and often requires that multiple and oblique innovation pathways be followed (Kay and King 2020).

Following the logic of the model proposed by Teece (2006, p. 1139) further, the next question is whether or not firms have the cash position required to integrate the specialized complementary assets, which for dominant digital platforms is generally the case. In such cases, the Teecean model would encourage digital platforms to integrate the specialized complementary assets themselves but only if imitators or competitors are not better positioned to command them. Yet, as will be shown in Section 3, the logic underlying this decision is much more subtle and complex in the case of digital platforms and generally contradicts the logic of the Teecean model.

In the next section, we will describe several cases of innovation ecosystem extension. The first one corresponds to the case of Twitter after its successful IPO in 2013, which aimed at finding a viable revenue model. We will then introduce the more serendipitous case of innovation ecosystem extension of Facebook with Cambridge Analytica and the case of innovation ecosystem extension executed by Google in the online devices and services industry, which was predicated on its mobile operating system Android and attempted to dethrone the iOS mobile operating system and the Apple Store as the incumbent digital platform in that industry. We will also introduce the new grand strategy of Google around Alphabet, which showcases how incumbent digital platforms are currently disrupting existing industry sectors and creating new industries from scratch.

[5] These cases, which are by far more common due to the myriad of ways in which big data can be monetized and the deep uncertainty associated with developing the necessary assets for such monetization, are best exemplified by the "Moonshots" and "Other Bets" entrepreneurial agendas driven as part of the new grand strategy of Google (Manjoo 2015),

Our analysis of these cases will reveal how the process of innovation ecosystem extension takes place and will also show how incumbent digital platforms with a solid cash position and strong innovation capabilities will often engage with interested parties in their extended innovation ecosystems and let them develop the cospecialized assets these incumbents need to profit from innovation, even at the risk that these interested parties may turn into competitors in the future.

3 Cases of innovation ecosystem extension

We begin this section by reviewing the case of innovation ecosystem extension executed by Twitter.

3.1 The search for a revenue model in the extended ecosystem of Twitter

Launched as a double-sided platform, Twitter created the microblogging market with two types of users, of which the followers where the more numerous group. Unlike other digital platforms, the content generated by Twitter's users was not only public but also highly time-sensitive. This allowed Twitter to differentiate itself from other dominant platforms such as Google and Facebook. The result was a rapid growth to almost 300 million active users at the time of its IPO in 2013 (Welch and Popper 2015).

Adopting the revenue model of Google and Facebook based on computational advertising turned out to be not a viable option for Twitter and the company struggled after its successful IPO to find a viable revenue model. Twitter's value proposition to advertisers based on selling Internet ads based on micro blogs was simply not compelling enough and Twitter could not compete with incumbent digital platforms such as Google and Facebook (Courtright 2019). Here is where the strategy of Twitter to build an extended ecosystem comes in.

Leveraging the value creation potential of its big data to deploy bespoke services based on data analytics but lacking the needed resources, capabilities and competencies to explore all possibilities in parallel in the face of deep uncertainty, Twitter first formed an extended ecosystem by granting interested parties access to the tweets posted, either offline through its search API or in near realtime through its streaming API. Both options gave access to a small percentage of all tweets matching a query and were offered free of charge to users. This enabled Twitter to form an ecosystem of interested parties comprised of highly innovative firms developing cospecialized resources, capabilities and competencies with the aim of launching bespoke services in a variety of niche application domains based on technologies ranging from data analytics to Internet sentiment analysis (Weller et al. 2014). Twitter also experimented with a revenue model based on Twitter's Firehose, which gave access to all tweets matching a search criterion for a fee, a business handled by a firm that Twitter ended up acquiring in 2014 (Ingram 2011).

A central element of this strategy was the positioning of Twitter's big data as a generalized complementary asset in its extended innovation ecosystem. This strategy allowed Twitter to entice a great number of interested parties to develop cospecialized resources, capabilities and competencies in key technologies, ranging from data analytics, machine learning and user modeling to Internet sentiment analysis and natural language processing. Twitter's strategy consisted in deploying external resources, capabilities and competencies and explore multiple innovation pathways in parallel in order to find the cospecialized assets needed to profit from its big data. The success of this ecosystem extension strategy was evidenced by the large number of alliances Twitter entered into and the acquisitions it later executed, which according to AcquiredBy added up to over 50 as of October of 2018.[6]

3.2 The extended ecosystem of Facebook

The case of Facebook with Cambridge Analytica is an example of an interested party that became a stakeholder in the extended ecosystem of Facebook in a rather serendipitous way, as opposed to the strategic approach to the formation of a comprehensive extended innovation ecosystem followed by Twitter. In this second case, the innovation ecosystem of Facebook "got extended" by Cambridge Analytica, an interested party that was able to get access to user profiles of Facebook in order to provide a very special form of market intelligence information, namely, voter-profiling information (Cadwalladr and Graham-Harrison 2018).

This ad-hoc extension of its innovation ecosystem backfired on Facebook due to ethical issues regarding (lack of) user data protection. Yet, more than misappropriation of user data, the problem with this extension of Facebook's innovation ecosystem was its complete lack of strategic alignment with the objective and mission of Facebook, which had been changed to "building community and bringing the world closer together" in 2017. The rationale for ecosystem extension in this case can be characterized as business model innovation by exploring and finding new uses for the big data of an incumbent digital platform. Finding such new uses entails a process of assigning new meanings to the big data of incumbent digital platforms and integrating such new uses to deliver value to new clients (Devlin 1999). The voter-profiling service envisioned by Cambridge Analytica corresponded indeed to new uses that could be offered to corporate clients and individuals. This case shows how incumbent digital platforms can commit strategic follies of colossal proportions when executing strategies of ecosystem extension.

Due to the strategic uncertainty that is inherent to the new uses that are explored and then exploited, either by the digital platforms themselves or by interested parties such as Cambridge Analytica in the case of Facebook, the risk of committing such strategic follies usually looms large and can pose an immediate threat to digital platforms executing strategies of business model innovation predicated on the extension of their existing innovation ecosystems. In such cases, immediate action is required

[6] The complete list is available at https://acquiredby.co/twitter-acquisitions.

to eliminate this threat. Facebook is a good example of such a reaction, as revealed by the suspension of thousands of third-party applications, a measure that was executed by Facebook right after the Cambridge Analytica scandal.[7]

3.3 Google's innovation ecosystem extension strategies

An early prototypical case of innovation ecosystem extension was delivered by Google's attempt to dethrone Apple's iOS as the leading mobile operating system with its own Google Play platform. This case is paradigmatic not only because Google provided Android as an open source platform to developers and as a mobile operating system to smartphone manufacturers. Google also renounced capturing profits and decided to transfer them to interested parties in order to maximize value creation in its extended innovation ecosystem around Android and Google Play. Launched in 2010, this "catch-up strategy" was extraordinarily successful and resulted in the number of applications available on Google Play exceeding those available on the Apple Store by 2014 (Parker et al. 2016).

A more comprehensive case of innovation ecosystem extension strategies, and one that is showcasing the evolution other incumbent platforms will undergo in the future, is Google's new grand strategy. This strategy involves several innovation ecosystem extension strategies under Alphabet as the new corporate umbrella founded on October 2, 2015, of which Google is now only one of a myriad of firms at different stages of development (Manjoo 2015). Although these initiatives had been under development for quite some time within Google and benefited from scope economies based on some of Google's core and complementary assets,[8] there were several reasons why Google decided to spin out several of its intrapreneurship initiatives as separate firms under a new corporate umbrella called Alphabet.

Many of the intrapreneurial agendas driven by Google were at early stages of development and were thus subject to deep strategic uncertainty. They were not aligned with Google's core business and lacked the cospecialized complementary assets needed to turn them into profitable businesses (Zenger 2015). As a public company, Google also lacked the flexibility and agility needed to execute strategies aimed at creating new industries from scratch, which represented much more speculative innovation agendas and were thus referred to as "Moonshots" (Ward 2018).

Google's new grand strategy also called for an extension of its innovation ecosystem due to the threats posed by mobile devices and virtual assistants to Google's core search engine marketing business and to some of its other "Alpha Bets." This new grand strategy also gave the founders of Google the opportunity to execute diversification strategies in mature industries where Google lacked strategic and organizational alignment, such as the logistics and health industries, which were part of what was referred to as the "Other Bets" within Google.

[7] Facebook seems to have learned this lesson the hard way as a result of the Cambridge Analytica scandal (see www.nytimes.com/2019/09/20/technology/facebook-data-privacy-suspension.html).

[8] Such as Google's cutting-edge AI technologies and the sheer amount of big data harvested by Google's search engine marketing business.

As a publicly traded company, the successful execution of these entrepreneurial agendas was compromised due to a lack of strategic alignment with the short-term objectives of Google's shareholders. The founders understood that the needed organizational alignment of the firm to achieve the different goals that each of these initiatives entailed was difficult to accomplish under the old corporate structure. The corporate reorganization of Google was designed to endow Alphabet with the flexibility needed to cope with this challenge.[9]

Strategies of innovation ecosystems extension are already being deployed by incumbent digital platforms in order to deal with deep strategic uncertainty (Kay and King 2020). In the case of Twitter, the goal was to find a viable revenue model and render Twitter profitable, an objective that proved elusive for many years after its IPO in 2013 (Morgan 2015). In the case of Facebook, the serendipitous innovation ecosystem extension strategy executed by Cambridge Analytica brought with it deep strategic uncertainty and lack of organizational alignment, which resulted in governance and business ethical issues that prevented Facebook from integrating Cambridge Analytica as a trusted partner able to diversify its business (Conger et al. 2019). In the case of Google's catch-up strategy, the goal was to dethrone iOS as the leading mobile operating system, which seemed like an unsurmountable task given the dominance of Apple in that space in 2010 (Parker et al. 2016). The most compelling case discussed in this section is by far the new grand strategy of Google with the creation of Alphabet. Indeed, the new grand strategy of Google is a prototypical case of innovation ecosystem extension strategies that will be executed by incumbent digital platforms in the future.

The cases analyzed in this section show the role that innovation ecosystem extension strategies will have in allowing digital platforms to gain the flexibility required to deal with the deep strategic uncertainty and the lack organizational alignment that will abound in their business environments going forward. In the next section, we introduce the model under which digital platforms operate today, which we have termed "the global stakeholder capitalism model of digital platforms."

4 The global stakeholder capitalism model of digital platforms

Corporations that operate under the global model of hypercompetition (D'Aveni 1994, 2010; D'Aveni et al. 2010) and the global stakeholder capitalism model (Hseih 2004; Nolan 2008) emerged as a result of deregulation and privatization of several industry sectors during the 1990s. Although they replaced the legacy business models of large state-owned monopolies, global corporations that operate under these two models continued to operate under the logic of linear value chains and are thus referred to as "pipeline businesses" (van Alstyne et al. 2006). They were also

[9] A comprehensive report on the new grand strategy of Google can be accessed at: https://www.cbinsights.com/research/report/google-strategy-teardown/.

subject to government regulation, which slowed down their global expansion. The new model under which digital platforms operate today constitutes quite a departure from these previous models for two main reasons.

The first reason is that digital platforms do not operate in linear value chains but in innovation ecosystems able to generate much stronger network effects. This allows them to benefit from the demand-side economies of scale of platform businesses as opposed to the supply-side economies of scale of conventional pipeline businesses. Based on the facilitation of interactions in group-forming networks, this new form of network effects, which can grow exponentially according to the law of the pack (Reed 2001), is absent in pipeline businesses.[10] The second reason is that the virtual markets created by digital platforms are completely unregulated. The effect of this lack of government regulation is further amplified by the positioning of the Internet as a global common or public good digital platform users can have access to without the need to deploy expensive network infrastructure globally. The result is the rise of the new global stakeholder capitalism model of digital platforms. Digital platforms operating under this new model have left the logic of pipeline businesses behind as a relic of the past and have been able to expand globally at a pace unprecedented in history.

In the next section, we discuss what makes this new model different from its predecessors.

4.1 Comparison with the global model of hypercompetition

Like firms under the global hypercompetition model (D'Aveni 1994, 2010; D'Aveni et al. 2010), digital platforms can also consolidate as global oligopolies. But digital platforms benefit not only from demand-side economies of scale that draw upon strong network effects in group-forming networks but also from lack of regulation. Once digital platforms reach a critical mass of users, they start commanding demand-side economies of scale and tend to take all or almost all the market in which they operate. Digital platforms can thus consolidate as global oligopolies much more straightforwardly. This begs the question of whether digital platforms are less vulnerable to the attack of new market entrants than the global oligopolies under the global model of hypercompetition.

Despite their significant market power, negative network effects can destroy the dominant position of incumbent digital platforms in the same way that positive network effects made them dominant in the first place, giving latent competitors a chance to take their place. The loss of Netflix of almost a million user accounts immediately after the firm changed its revenue model in 2011 is a paradigmatic case

[10] The rise of the online ad syndication market pioneered by Google at the turn of the millennium provided an early case of a digital platform harnessing the power of exponential network effects in group-forming networks to ignite a period of hyper growth. In the case of Google, this period gave rise to the search engine marketing industry as we know it today.

of how negative network effects can threaten to destroy the dominant position of a digital platform in a very short period of time (Wingfield and Stelter 2011). How Facebook replaced MySpace as the dominant social media networking application is another example of the same phenomenon that shows why digital platforms are indeed more vulnerable to the threat of new entrants than firms that operate under the global hypercompetition model (Jacobs 2015).

A key characteristic shared by digital platforms and firms under the global hypercompetition model is the difficulty to sustain a competitive advantage and the need to constantly innovate to obtain a temporary competitive advantage. But the way in which digital platforms drive innovation is fundamentally different. While firms under the global hypercompetition model compete by controlling both supply-chain platforms and the open innovation systems around them, digital platforms do so by orchestrating much more complex innovation ecosystems instead of controlling them.

4.2 Comparison with the global stakeholder capitalism model

Usually referred to as sides in the digital platform literature, the agents we find in the innovation ecosystems of digital platforms are much more heterogeneous than those we find in global firms that operate under the global stakeholder capitalism model. This heterogeneity makes the orchestration of these innovation ecosystems difficult. These agents also present much higher levels of multiplexity, as they can assume multiple roles and these roles can change and overlap over time. But what makes the global stakeholder capitalism model unique is not its focus on innovation but rather its adoption of one of the main political assumptions of the stakeholder capitalism model (Freeman 1984), namely, the rejection of the separation hypothesis of the classical liberal model (Friedman 1962).

The separation hypothesis makes a clear distinction between shareholders and all other stakeholders of the firm and gave rise to models of business ethics according to which good corporate governance should be based on the principle of maximizing shareholder value. By dropping the separation hypothesis, the global stakeholder capitalism model abandoned the strict distinction between shareholders and all other stakeholders of the firm and adopted a new form of corporate governance based upon Kantian business ethics (Bowie 1998). The global stakeholder capitalism model replaces the maxim of maximizing shareholder value of the classical liberal model with the maxim of optimizing stakeholder value (Freeman et al. 2007). As a result of dropping the separation hypothesis, firms under the global stakeholder capitalism model adopt a multi-fiduciary duty to "treat all stakeholders as ends in themselves, not purely instrumentally" (Singer 2013, p. 316).

The rise of a global stakeholder model in connection with digital platforms is a recent phenomenon that results from executing strategies of innovation ecosystem extension. The cases analyzed in Section 3 provide evidence that the execution of these strategies is already giving rise to extended innovation ecosystems comprised of stakeholders that go beyond the sides served by digital platforms.[11] This is rendering the governance of the extended innovation ecosystems of digital platforms very complex. Developments such as the Internet of Things and the Internet of Everything will bring a dramatic explosion of stakeholders in the extended innovation ecosystems of digital platforms and will make the governance of these ecosystems even more complex in the future.

Due to the complex governance of these extended innovation ecosystems and the need to ignite and orchestrate innovation in them, digital platforms will need to drop the separation hypothesis and adopt a multi-fiduciary duty to treat all sides and also all stakeholders in these ecosystems as ends in themselves. Whether such a multi-fiduciary duty under the global stakeholder capitalism model of digital platforms will be driven by a genuine need and desire to treat stakeholders as ends in themselves, as proposed by the global stakeholder capitalism model, or whether it will be purely instrumental and driven by narrow interests remains to be seen.

4.3 The rise of the global stakeholder capitalism model of digital platforms

Drawing upon demand-side scale economies and the lack of Internet regulation, successful digital platforms can ignite processes of hyper growth and take all or almost all the market in which they operate. As a result, they are in a privileged position not only to attain and exploit significant market power but also to extend it by engaging in strategic acquisitions of digital platforms that invade their space and can turn into potential competitors.[12] However, the market limitations exploited by digital platforms are not limited to monopolistic tendencies. They can include other known limitations of market-based systems, such as unpriced and unanticipated externalities,[13] as well as a whole new range of market limitations that will arise in connection with the global stakeholder capitalism model of digital platforms in the future.

But digital platforms are also in the privileged position to reduce market limitations through the reduction of information asymmetries in rigged markets and the elimination of the lack of ability to pay by providing free services to users. Digital platforms can also compensate for exploiting market limitations by complementing the role of governments in providing not only human goods (HGs) to society, such as "wealth, health, friendship, aesthetics, beauty and justice" (Singer 2013, p. 309), but also public goods (PGs) through direct investments in technological and social innovations.

The global stakeholder capitalism model of digital platforms is shown in Fig. 1, where the dark and light gray areas show the innovation ecosystem and the extended innovation ecosystem, respectively. The link from digital platforms to interested parties in Fig. 1 represents the provision of generalized complementary assets,

[11] The most ambitious case of innovation ecosystem extension is the new grand strategy of Google under Alphabet (Manjoo 2015).

[12] The acquisition of Instagram by Facebook is an example of this type of acquisitions.

[13] The sudden rise in price of property rentals in the case of Airbnb is an example of unanticipated externalities arising in the extended innovation ecosystem of digital platforms.

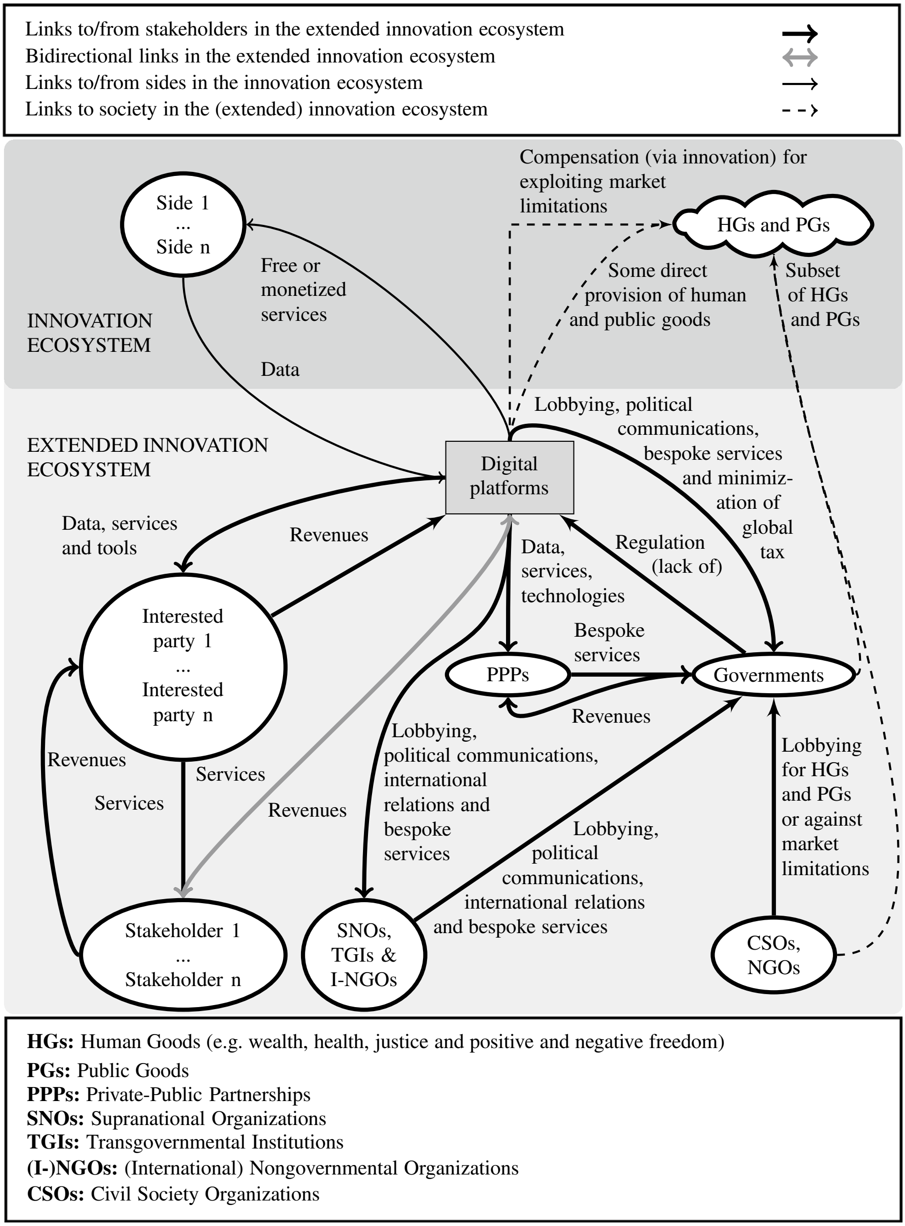

Fig. 1 The global stakeholder capitalism model of digital platforms

including big data, services and tools. The links to and from digital platforms and interested parties represent the situation in which digital platforms let interested parties explore new services and provide them to stakeholders in order to profit from them indirectly. The gray bidirectional link between digital platforms and the stakeholders represents the direct provision and monetization of services to stakeholders by digital platforms. This will be the case if digital platforms decide to explore new services themselves and are successful in finding a viable business model to profit from them or if they decide to disintermediate successful interested parties and integrate the cospecialized assets and business models developed by them. The latter will be the case if such services are of strategic importance or if they pose a threat to digital platforms. However, such integration strategies can be highly problematic, as they may send the wrong signal to other interested parties and can have a detrimental effect on the governance of the extended innovation ecosystem of digital platforms.[14]

Besides interested parties and stakeholders, the global stakeholder capitalism model of digital platforms also includes governments not only as policymakers and regulators but also as a strategic partners of digital platforms. As shown in Fig. 1, digital platforms can engage in lobbying and political communications with the aim of upholding corporate negative freedoms allowing them to exploit market limitations. Digital platforms are in a privileged position to provide bespoke services to governments under this model either directly or through private-public partnerships (PPPs). Such partnerships highlight the political and international relations dimensions of the construct underlying the global stakeholder capitalism model of digital platforms.

Digital platforms can also engage in lobbying, political communications and international relations not only with supranational organizations (SNOs) but also with transgovernmental institutions (TGIs) and international nongovernmental organizations (I-NGOs) and they can also partner and provide bespoke services to them. The relationships with these stakeholders either mediate or moderate international relations on a global scale and further highlight the international relations component of the construct underlying the global stakeholder capitalism model of digital platforms.

The global stakeholder capitalism model of digital platforms also includes civil society organizations (CSOs) and nongovernmental organizations (NGOs). As shown in Fig. 1, these organizations can play a role as a mechanism of digital governance by lobbying for the provision of human and public goods and/or against the exploitation of market limitations and they can also complement the role of governments in providing a subset of human and public goods to society.

4.4 The extended innovation ecosystems of digital platforms

The rise of digital platforms as global corporations has been radically different from the rise of corporations under the global hypercompetition model and the global stakeholder capitalism model in that digital platforms have implemented virtual micro and meso economies through the completely new logic of platform businesses. As global oligopolies, digital platforms share with the corporations that operate under

[14] The case of Twitter analyzed in Section 3 is a prototypical example of such integration process (https://www.vox.com/2015/4/11/11561360/twitter-cutting-ties-with-all-data-resellers-including-datasift).

the global hypercompetition model the ability to exploit significant market power. But unlike them, digital platforms can leverage the Internet as a global and unregulated infrastructure to rise as global oligopolies at a much faster pace.

Unlike digital platforms, the process of global consolidation of firms operating under the global stakeholder capitalism model and the global hypercompetition model does require that they engage ex-ante in political communications and lobbying with government bodies of the nation-states in which they operate. This makes their process of global consolidation anything but frictionless. Freed from legal and regulatory constraints, digital platforms may engage in political communications and lobbying only to keep the legal and regulatory status quo.

In the case of Twitter, the search for a revenue model was executed by extending the innovation ecosystem with interested parties that developed cospecialized complementary assets in as diverse areas as data analytics and Internet sentiment analysis. By providing big data as a generalized complementary asset to multiple interested parties in its extended innovation ecosystem, Twitter leveraged the resources, capabilities and competencies of these interested parties to explore the provision of new services to new groups of stakeholders in parallel. As a result, new stakeholders became part of Twitter's extended innovation ecosystem. But Twitter decided to disintermediate interested parties that had been successful in finding and exploiting new services and began to provide these services to stakeholders directly (Ingram 2011).

Such integration of the cospecialized complementary assets will often result in these new stakeholders becoming new sides in the innovation ecosystem of digital platforms. In other cases the interested parties and their stakeholders will remain in the extended innovation ecosystem. The case of Facebook and Cambridge Analytica, if successful, would have been such a case, as the lack of strategic alignment between both companies would have made an integration of the cospecialized assets developed by Cambridge Analytica undesirable. But even in cases where such an integration were desirable, digital platforms would often refrain from it. Google's catch-up strategy is a wonderful example.

Google's catch-up strategy included as interested parties not only handset manufacturers that competed with Apple, some of them possessing their own mobile operating system, but also developers, including applications developers. Google's plan was to allow developers to add functionality to Android and to piggyback on Google Play to launch their own mobile applications. Without a critical mass of Android users to attract the attention of these interested parties in 2010, this strategy was ridden with deep strategic uncertainty. Against all odds, however, this strategy proved successful and resulted in the total number of applications running on Google Play exceeding those running on the Apple Store in less than four years (Parker et al. 2016).

While the cases of Twitter and Google mentioned above provide us with early examples of strategies of innovation ecosystem extension, the new grand strategy of innovation ecosystem extension of Google under its new corporate umbrella Alphabet is the most complex and comprehensive case of innovation ecosystem extension to date.

In the next section, we discuss the implications of the global stakeholder capitalism model of digital platforms for strategy and innovation.

5 Discussion

Digital platforms operating under the model described in Section 4 orchestrate innovation ecosystems that are already complex. But the extended ecosystems that digital platforms will enable in the future will become even more complex due to the higher heterogeneity and multiplexity of the stakeholders that will be involved in them. We begin our analysis in this section by discussing the implications this added complexity of the global stakeholder capitalism model of digital platforms have for innovation from a Schumpeterian perspective.

5.1 Implications for innovation

The shift away from the entrepreneur and toward the innovation networks in which she/he is embedded is not new in neo-Schumpeterian frameworks. This begs the question of whether the model of digital platforms has an impact on these frameworks in any fundamental way. We think that it does for the reasons we elaborate upon below.

5.1.1 From innovation networks to extended innovation ecosystems

Neo-Schumpeterian frameworks have long departed from their original view of the entrepreneur as the driving force behind innovation, as originally assumed by Schumpeter (1911). These frameworks highlight the innovation networks in which the entrepreneur is embedded as a key driver of innovation and creative destruction. Although neo-Schumpeterian frameworks acknowledge the important role that innovation networks play in driving innovation and creative destruction (Hanusch and Pyka 2007), the key role that the embedding function in complex innovation networks plays in innovation and entrepreneurship has been made more explicit only in the field of economic sociology. Mark Granovetter along with other economic sociologists started to write about the problem of embeddedness since the mid 1980s and the study of this problem has now taken center stage in economic sociology (Granovetter 1985; Castilla et al. 2000; Granovetter 2005; Powell et al. 2005; Ferrary and Granovetter 2009).

As shown in Fig. 1, the extended innovation systems of digital platforms go beyond the orchestration of open innovation systems to also encompass the orchestration of extended innovation ecosystems, which can be best understood as political subsystems embedding new interested parties that span across the political systems of nation-states. The orchestration of these extended innovation ecosystems is a novel and important function of digital platforms that we discuss in Section 5.3. Digital platforms also fundamentally change the way in which firms address the problem of value creation and value capture, which leads not only to a reconceptualization of the rents digital platforms capture but also to important changes to the standard model of how firms profit from innovation (Teece 1986, 2006).

5.1.2 Reconceptualizing Schumpeterian rents

Digital platforms are rapidly contributing to a reconceptualization of strategy and how firms drive and profit from innovation. This reconceptualization has also important implications for the generally accepted notion that innovative firms need to capture Schumpeterian rents while their competitors are not yet able to catch up with their innovations. The new global stakeholder capitalism model of digital platforms challenges the well-established concepts of Schumpeterian innovation, imitation and competition and calls the maxim of "capturing Schumpeterian rents while your competitors try to imitate you" into question. A new strategy of transferring Schumpeterian rents outside the firm to cocreate value in the extended innovation ecosystems of digital platforms is at play and becomes more relevant under this new model.

The new global stakeholder capitalism model of digital platforms challenges conventional strategic management theories rooted in the resource-based view of the firm (Barney 1991; Penrose 1995), which focus on driving competitive advantage through owning and controlling VRIN resources.[15] This model also challenges the positioning framework of strategic management (Porter 2008), which focus on driving competitive advantage and capturing Porterian rents by raising structural barriers to entry and increasing bargaining power in an industry sector.

The new global stakeholder capitalism model of digital platforms also calls for a revision of the concept of Schumpeterian rents. Instead of capturing Schumpeterian rents in disequilibrium, this model often relies on transferring most of the rents generated through innovations orchestrated by digital platforms to stakeholders in their extended ecosystems. This leads to a new concept of rents that neither corresponds to Schumpeterian or Kirznerian rents in disequilibrium nor to the kind of Porterian rents in equilibrium advocated by mainstream strategic management frameworks rooted in neoclassical economics (Porter 2008).

The very idea of renouncing the lion's share of profits generated via innovation by transferring this new form of rents to stakeholders in the extended innovation ecosystem of digital platforms has not yet been accommodated by more modern strategic management frameworks rooted in the resource-based view of the firm, such as the dynamic capabilities framework (Teece 2014), and runs counter to the standard view of how firms profit from innovation (Teece 1986, 2006, 2018a, b). We argue that this new form of rents, which we might term "digital rents under the new global stakeholder capitalism model of digital platforms," constitutes the key pillar upon which the strategy of digital platforms rests.

5.1.3 The dilemma of innovation ecosystem extension

When extending their innovation ecosystem, digital platforms have the dilemma of whether or not to integrate the new services offered by interested parties once they have been successful not only in developing the cospecialized assets needed but also in finding a viable revenue model to offer these new services and profit from them. We refer to this dilemma as the "dilemma of innovation ecosystem extension."

[15] These are valuable, rare, inimitable and nonsubstitutable resources.

The decision to stop providing big data to interested parties as a generalized complementary asset and integrate these new services, along with the cospecialized assets that are needed to profit from them, is always difficult to make because it can have negative repercussions with other interested parties in the extended innovation ecosystems of digital platforms. But in some cases this course of action is necessary.

In order to nurture cooperation in their extended innovation ecosystems, digital platforms should in general refrain from integrating these cospecialized assets and destroy the business model of successful interested parties that are highly dependent on the steady supply of big data as a generalized complementary asset. But such strategy of integration is warranted in some cases, especially if the interested party involved threatens to interpose itself between the platform and its users and installs what Parker et al. (2016) refer to as "user control points." Faced with this dilemma, Twitter severed the alliance with interested parties that had been successful in cocreating value in its extended innovation ecosystem and invaded their space by integrating their cospecialized assets.[16] Interested parties whose business models depend on the supply of big data from incumbent digital platforms face in such cases the destruction of their business models (Ingram 2011; Morgan 2015). Incumbent digital platforms that decide to resolve this dilemma by pursuing the cooperate-to-then-absorb strategy often do so violating rules of digital platform governance (Gawer and Cusumano 2002; Cusumano et al. 2019).

Following the logic of the Teecean model, digital platforms may be prompted to integrate the complementary assets provided by interested parties. But such actions on the part of digital platforms may often backfire on them. As they may send confusing signals to other interested parties, the strategy of integration can destroy the goodwill of interested parties in the extended innovation ecosystems of digital platforms very quickly and can ruin a reputation that took digital platforms years to build (Lunden 2015; Morgan 2015).

5.1.4 The extended governance of digital platforms

The rise of the extended ecosystems of digital platforms is making the governance of digital platforms transition from governing the sides served by the platform to governing the stakeholders that are served by the platform either directly or through interested parties. This is due to the higher heterogeneity and multiplexity of the stakeholders in the extended innovation ecosystems of digital platforms.

From a neo-Schumpeterian perspective, the resolution of this new problem of extended governance will take center stage and become the key strategic pillar on which not only the innovation strategy of digital platforms but also the preservation of the global stakeholder capitalism model of digital platforms will depend in the future. The high heterogeneity and multiplexity of the stakeholders that will be found in the extended innovation ecosystems of digital platforms will lead to a myriad of complex trade-offs. As more stakeholders from the public and private sector join the extended

[16] See for example https://www.vox.com/2015/4/11/11561360/twitter-cutting-ties-with-all-data-resellers-including-datasift).

innovation ecosystems of digital platforms as interested parties,[17] these trade-offs will not only be more numerous but also more complex. The resolution of these complex trade-offs through a new form of "extended governance" will be essential to preserve the global stakeholder capitalism model of digital platforms in the future. In order to cope with this formidable challenge, neo-Schumpeterian approaches need to be extended with integrative models of stakeholder governance at the intersection of strategy, business ethics, politics and international relations.

5.1.5 Integrative models of digital governance

The governance of digital platforms will require the implementation of evaluation procedures involving multiple and often conflicting criteria as well as processes of group decision and negotiation for social choice (Fishburn 1973; McLean and Urken 1995). Different stakeholders, most notably civil society, are going to exert increasing pressure on dominant digital platforms to deliver social welfare in the form of social innovations diffused in the extended innovation ecosystems of digital platforms. As a result of such pressures, digital platforms will need to master the hard balancing act between exploiting the limitations of the markets they currently serve, and especially the limitations of the market-based systems they will create and serve in the future, on the one hand, and delivering social innovations in their extended innovation ecosystems and society at large, on the other. This task will become all the more difficult to accomplish as dominant digital platforms disrupt and gain significant market power in new industry sectors.

The global stakeholder capitalism model of digital platforms has also important implications for the standard model of how firms profit from innovation. We elaborate on some of these implications below.

5.2 Revisiting Teece's "profiting from innovation" framework

The Teecean model introduced in Section 2 is considered the standard model of how firms profit from innovation (Teece 1986, 2006, 2018a, b). In scenarios in which innovations are subject to weak appropriability regimes and the innovator "is disadvantageously positioned vis-à-vis independent owners of complementary assets" and "is excellently positioned versus imitators with respect to commissioning complementary assets,"[18] the Teecean model dictates that the innovator "should integrate the complementary assets" needed to profit from innovation and predicts that "the innovator should win" (Teece 1986, p. 297). However, such course of action may often constitute a strategic folly of colossal proportions for digital platforms. In fact, dominant digital platforms often provide big data as a generalized complementary asset, as well as other core and complementary assets such as access to users and technology, to other interested parties in their innovation ecosystems in a way that was

[17] A recent example of these complex trade-offs is the current debate on how digital platforms should regulate fake news without limiting free speech.

[18] Which corresponds to cell 5 in the original diagram 11 introduced in Teece (1986, p. 297).

neither foreseen in the original Teecean model of how firms profit from innovation (Teece 1986) nor addressed in its subsequent revisions (Teece 2006, 2018a, b).

The unit of analysis in Teece's original article was that of an innovation understood as "certain technological knowledge about how to do things better than the existing state of the art" (Teece 1986, p. 288). However, the core asset that fuels innovation in the case of digital platforms is not technology but big data, which confronts digital platforms with a difficult choice between two seemingly opposite strategies. The first strategy not only consists in positioning big data as a core asset that is kept captive to the platform's own services but also in integrating the needed complementary assets to gain significant market power. The second consists in positioning big data as a generalized complementary asset that is made available to other interested parties, allowing them to drive innovation in the extended innovation ecosystems of digital platforms and profit from it. The resolution of this dilemma is complex and is analyzed below.

5.2.1 Complementary assets

From the perspective of digital platforms, big data is a core resource that can enable a myriad of possible innovations. But complementary assets that are not freely available in the ecosystems of digital platforms are needed in order to unleash the value of big data and drive these innovations. Digital platforms have the option to either develop these complementary assets themselves or contractually engage with interested parties in their innovation ecosystems to have access to such complementary assets.

Seen from the perspective of those external interested parties, the process of innovation is driven by them through the development, ownership and control of highly cospecialized assets. To these interested parties, such cospecialized assets are not complementary but core assets needed to develop and profit from their innovations. Crucially, the innovations driven by these interested parties require that they develop cospecialized assets based on big data provided to them as a generalized complementary asset by digital platforms.

5.2.2 Appropriability regimes

We begin this section with an analysis of tight appropriability regimes.

Tight appropriability regimes In the Teecean model, firms that benefit from tight appropriability regimes can afford to subcontract for the development of the complementary assets they need due to the "inimitability" of their innovations. As stated in its original article, "competition from imitators is muted in such appropriability regimes" (Teece 1986, p. 290). Hence, the innovating firm is in this case advised not to develop the complementary assets it needs in the preparadigmatic phase as it can rely on subcontractors to develop them. But in case these complementary assets were specialized or cospecialized, then the Teecean model recommends that such assets be integrated, as any contractual agreement between digital platforms and the interested parties involved would be "exposed to hazards and could break down" (Teece 1986, p. 290). Accordingly, digital platforms should avoid this hazard by refraining from extending their innovation ecosystems in order to let interested parties develop the cospecialized assets needed. Yet digital platforms often have to do that. However, tight appropriability regimes are the exception rather than the rule in the case of digital platforms. This leads us to consider the more prevalent situation in which digital platforms and their innovations are subject to weak appropriability regimes.

Weak appropriability regimes Digital platforms innovate under appropriability regimes that are weak or ineffective in terms of providing real protection for their innovations, as best exemplified by the ongoing legal battles between Apple and Samsung (Cain 2020). Lacking efficient markets providing the complementary assets needed to profit from innovation, digital platforms are often forced to create innovation ecosystems comprised of heterogenous stakeholders. And once these ecosystems are formed, they need to identify key stakeholders and entice them to develop the cospecialized assets that are needed to drive innovation.[19] However, this does not contribute to the formation of efficient markets for complementary assets but rather to the creation of firms in the extended innovation ecosystems of digital platforms that end up exerting tight control over these cospecialized assets and may turn into competitors.

Under weak appropriability regimes, the Teecean model would dictate that digital platforms should develop and keep the core and complementary assets needed to profit from their innovations captive to their own future services (i) if these assets are not yet available internally; (ii) if they are specialized and critical; (iii) if digital platforms have a strong cash position to develop them; and (iv) if competitors or imitators are not better positioned to develop them (Teece 1986, p. 296). Yet digital platforms often have to extend their innovation ecosystems and engage interested parties to develop the needed cospecialized assets when these conditions apply.

Initially, digital platforms create and extend their innovation ecosystems by granting interested parties access not only to their platforms but also to their big data. Interested parties that commit to developing the cospecialized assets needed to drive innovation in these extended innovation ecosystems have to rely on principles of good governance on the part of digital platforms (Gawer and Cusumano 2002). Interested parties also face the ex-ante possibility of failure and losses if they are unsuccessful in developing the needed cospecialized assets. But even if they are successful, they still face the possibility that their relationship with the incumbent digital platform breaks down and their business gets absorbed by them. Unfortunately for them, such an outcome[20] is not an unlikely one.

Yet once the interested parties have not only developed the cospecialized assets but also begun to profit from them, this asymmetrical relationship can be turned around and result in a threat to digital platforms. This is often the case when interested parties that launched their applications by piggybacking on incumbent digital platforms reach a critical mass of users. This is a potential threat to incumbent

[19] This process is best exemplified by Twitter in its quest for a viable revenue model (Welch and Popper 2015).

[20] As exemplified by Twitter's governance of its innovation ecosystem (Ingram 2011).

digital platforms that can turn into a strategic trap when the applications of these interested parties become platforms in their own right catering to the same (or substantially the same) users served by the incumbent digital platforms they piggybacked on (Eisenmann et al. 2007).[21]

While these asymmetrical relationships are and remain inherently hazardous, incumbent digital platforms often need to engage in them as a matter of necessity to drive innovation in their innovation ecosystems. Keeping captive the core and complementary assets needed to profit from innovation, as dictated by Teecean model, will often turn out to be a strategic folly for digital platforms, as the emblematic case of MySpace versus Facebook exemplifies (Rosenbush 2006; Jacobs 2015).

5.2.3 Integrating complementary assets in the preparadigmatic phase

Even if their market power allows digital platforms to protect their core assets, this protection is fleeting and can only be achieved temporarily (D'Aveni 1994, 2010; D'Aveni et al. 2010). From the perspective of digital platforms, the cospecialized assets owned and controlled by interested parties are assets that are needed to profit from innovation. And contrary to the tenets of the Teecean model, the integration of such cospecialized assets is often a strategic folly that digital platforms should avoid in the preparadigmatic phase. The reason for this lies in the deep uncertainty that pervades the conversion of big data into information, defined as data with meaning, and ultimately into new knowledge, defined as information for which new valuable uses have been discovered and integrated by a group of stakeholders (Devlin 1999).

The conversion of big data into information and into knowledge presupposes a process of exploration and discovery that is ridden with deep uncertainty. While digital platforms may be tempted to engage in such a process in stand-alone mode in order to retain control, there are many reasons why they should not pursue this course of action. Chief among them is that letting interested parties do the exploration and discovery on their behalf allows digital platform owners to deal with deep uncertainty by leveraging resources and capabilities that lie outside their platforms.

This strategy allows digital platforms to pursue multiple innovation pathways in parallel and dramatically increases the chances of success in finding new uses for big data. It also maximizes the innovations that digital platforms can diffuse in their extended innovation ecosystems, which are the ultimate source of their competitive advantage.

5.2.4 Integrating complementary assets in the paradigmatic phase

Applying the logic of the Teecean model, digital platforms should refrain from engaging with interested parties and should integrate the cospecialized assets needed to profit from their innovations if the conditions listed in Section 5.2.2 hold true (Teece 1986, p. 296). This is the strategy that MySpace followed, which caused

[21] An early case of this threat was delivered by YouTube as a highly successful application that piggybacked on MySpace (Rosenbush 2006).

its demise and opened up the market for Facebook to take its place as the dominant social media platform in the early 2000s (Arango 2011; Jacobs 2015).

As shown in Section 3, Twitter did not follow the logic of the Teecean model in the paradigmatic phase and executed the exact opposite strategy of MySpace by letting armies of interested parties in its extended innovation ecosystem find and develop the cospecialized complementary assets needed to profit from the big data delivered by Twitter's microblogging application, which had already entered the paradigmatic phase and had no competitors.

These and other cases of digital platform failure and success strongly suggest that the Teecean model also collapses in the paradigmatic phase in the case of digital platforms. This begs the question of whether digital platforms should severe their contractual relationships with interested parties in the paradigmatic phase after having benefited from such relationships in the preparadigmatic phase. In other words, should incumbent digital platforms severe these relationships, integrate the cospecialized assets, and adopt the viable and profitable business model of those interested parties that succeeded and showed them the way in the preparadigmatic phase?

Following the strategy of integration, digital platforms should use their asymmetrical relationship while it lasts to bring in house the cospecialized assets in question, replicate the business models of successful interested parties in their extended ecosystems, and perfect them via business model innovation. Although not covered by the original Teecean model, such a "cooperate-to-then-absorb" innovation ecosystem extension strategy would be coherent with the logic of the Teecean model. However, as some of the cases analyzed in Section 3 show (Rosenbush 2006; Ingram 2011), such a course of action could seriously compromise the sustainability of digital platforms and erode their main source of competitive advantage, namely, the value delivered by the innovations they diffuse in their extended innovation ecosystems.

While dominant platforms can indeed execute the "cooperate-to-then-absorb" strategy once the strategic uncertainty in their extended ecosystems has been reduced, they should do so only in extreme cases, that is, only when the interested parties became a real threat to the sustainability of their business models. This could be done by either imitating the cospecialized assets of interested parties in the paradigmatic phase or by way of an outright acquisition of the interested parties involved. Although acquisitions of unrelated digital platforms are common, acquisitions of interested parties in the extended innovation ecosystems of digital platforms are often problematic due to the strategic trade-offs they usually entail.

While the strategic acquisition of unrelated platforms by incumbent platforms, such as the acquisition of WhatsApp by Facebook (Olson 2018), tends to be more straightforward from a governance standpoint, the severance of strategic alliances with interested parties in the extended innovation ecosystems of digital platforms often poses governance issues, especially in cases that involve several interested parties. The acquisition of YouTube by Google, which also involved MySpace, the platform YouTube was still piggybacking on at the time of the acquisition, is a prototypical example of such cases (Rosenbush 2006). As far as MySpace was concerned, the acquisition of YouTube by Google was a case of the so-called "dilemma of innovation ecosystem extension," which we introduced in Section 5.1.3.

5.2.5 Profiting from innovation in the digital economy

David Teece recently outlined some extensions to his model to better capture the way digital platforms profit from innovation (Teece 2018a, p. 1382-1383). Below, we discuss these extensions from the perspective of our model.

Licensing and underinvestment in enabling technologies Owing to the focus of the Teecean model on value capture as opposed to value creation (Teece 2018b, p. 1400), it is argued that relying on licensing, as the revenue model to profit from innovation in enabling technologies, does not capture the profits needed to provide the necessary incentives to invest in the development of enabling technologies and therefore constitutes a serious problem.

The positioning of big data as a generalized complementary asset goes beyond usual licensing schemes. We argue that the situation of dominant digital platforms, construed not as internal or supply-chain platforms but as industry platforms that drive innovation in innovation ecosystems (Gawer 2014), are not affected by this problem. Even incumbent digital platforms that are able to secure a dominant position and exert significant market power still operate under global hypercompetition, which makes the development of not only enabling technologies but also general purpose technology platforms essential to secure a temporary competitive advantage (D'Aveni 1994, 2010; D'Aveni et al. 2010).

The pressure to continually redefine and transform the sources of competitive advantage is much higher for digital platforms due to the nature of the innovation ecosystems in which they operate. As these ecosystems are orchestrated rather than controlled by digital platforms, latent competitors can arise from the most unexpected quarters and overtake the dominant position of an incumbent platform in a relatively short time by drawing upon the same kind of strong network effects that made the dominant digital platform successful. Keeping this threat at bay requires the development of enabling technologies and general purpose technology platforms as sources of competitive advantage during the preparadigmatic phase despite the possibility of failure and losses.

Lack of market mechanisms to invest in enabling technologies The third revision of the Teecean model proposes that "there are no market mechanisms to ensure sufficient investment in the enabling technologies society needs ... and innovators can not solve the bargaining problem due to the inherent difficulty of identifying the recombinant possibilities ex ante and contracting so as to achieve the needed coordination." Yet no mechanisms are proposed or suggested.

Some of the cases analyzed in Section 3 show how digital platforms make such investments in enabling technologies during stages of the preparadigmatic phase that are ridden with deep strategic uncertainty. During this phase, neither the "recombinant possibilities" nor the revenue and business models needed to profit from innovation can be anticipated. In order to deal with the deep strategic uncertainty that is inherent to this process of exploration, digital platforms often engage interested parties to deploy resources, capabilities and competencies that lie in their extended innovation ecosystems. This has the advantage of allowing them to explore multiple innovation pathways. This strategy may be more often deployed to drive the exploration and development of enabling technologies rather than general purpose technologies, as the latter require more control and coordination.

By positioning big data as a generalized complementary asset in their extended innovation ecosystems, digital platforms generate a strong superpositioning effect that allows them to explore several innovation pathways simultaneously. This process unfolds indirectly and requires that digital platforms entice interested parties to deploy resources, capabilities and competencies to explore and discover new offerings. Instead of "contracting to achieve the needed coordination" (Teece 2018a, p. 1382), digital platforms orchestrate the activities of interested parties and stakeholders in their extended innovation ecosystems by providing them with the necessary incentives to profit from their innovations. This will often require that digital platforms renounce the lion's share of profits to the interested parties involved in a blatant violation of some of the key tenets of the original Teecean model of how firms profit from innovation and its subsequent revisions (Teece 1986, 2006, 2018a, b).

Lack of public policies fostering and enforcing strong appropriability regimes Another area highlighted by the third revision of the Teecean model is the need for public policies fostering and enforcing strong appropriability regimes so as to ensure the innovative firms capture "more than a modest amount of the upside associated with enabling and general purpose technologies" (Teece 2018a, p. 1382).

Our response to this challenge is not to wait for the regulator to enact such public policies. This third revision of the Teecean model misses entirely a key point that has been made evident by digital platforms scholars, namely, that digital platforms can act a regulators and policymakers themselves, and that they can enact public policy in several areas including fiscal and monetary policy, industrial policy, foreign policy and industrial policy (Lessig 2006). The case of Twitter analyzed in Section 3 is an example of how a digital platform can enact industrial and IP policy in order to promote the development of enabling technologies.

Owing to the importance of the big data they generate and contribute to interested parties in their extended innovation ecosystems, digital platforms have a strong bargaining position that is more effective than traditional appropriability regimes based on patents and other forms of IP protection. Instead of waiting for governments and regulatory agencies to install regulation enabling digital platforms to develop enabling and general purpose technologies, the digital platforms themselves are in the privileged position to act as policymakers in their extended innovation ecosystems, enticing interested parties to codevelop enabling and general purpose technologies and the complementary and cospecialzed assets needed to drive and profit from innovation.

Dynamic capabilities to profit from innovation Besides developing enabling and general purpose technologies, the third revision of the Teecean model highlights the importance of deploying dynamic capabilities for value capture. Unfortunately, the dynamic capabilities in question were not specified (Teece 2014).

Digital platforms can let interested parties and stakeholders in their extended innovation ecosystems drive the development of enabling and general purpose technologies and develop new business models for value capture by giving them the necessary incentives to explore multiple innovation pathways in parallel as an effective way to deal with radical uncertainty (Kay and King 2020).

The key point is that while platforms need to deploy dynamic capabilities as orchestrators of their extended innovation ecosystems, the interested parties and stakeholders in these ecosystems can also deploy a wide range of dynamic capabilities to be effective in the process of exploration of the cospecialized and complementary assets they need to profit from innovation. This has a compounding effect that can accelerate the deployment of the dynamic capabilities needed not only for sensing and seizing new service offerings but also for reconfiguring and transforming the business models needed to monetize them.

Multi-invention contexts The third revision of the Teecean model also highlights the need of firms to operate in multi-invention contexts. As they are more numerous and heterogeneous, the stakeholders in the extended innovation ecosystems can help digital platforms drive innovation in multi-invention contexts. The new grand strategy of Google is a quintessential example of how these extended innovation ecosystems can allow digital platforms to achieve this goal.

Business ecosystems The third revision of the Teecean model also highlights "business ecosystems as the relevant competitive arena for assessing profit-from-innovation issues." While we concur with this statement, we argue that the focus on value capture as opposed to value creation of the Teecean model can often backfire on dominant digital platforms. The catch-up strategy of Google described in Section 3.3 is a quintessential example of how digital platforms often need to focus on value creation in its innovation ecosystem and not on value capture. This focus on value creation allowed Google to implement a successful strategy that resulted in dethroning Apple's iOS as the incumbent mobile operating system.

From the perspective of the new strategy of digital platforms, the focus of the Teecean model on value capture over value creation appears to be a relic of the past that can lead dominant digital platforms to commit strategies follies of colossal proportions. The primordial case of MySpace mentioned in Sections 4.1 and 5.2.2 is a classic example of a strategic folly that costed MySpace its leadership at the turn of the millennium and paved the way for Facebook to take its place.

Differentiating complementarities The third revision of the Teecean model identifies a taxonomy of six categories of complementarities, two of which are more relevant for innovation. However, a taxonomy of the complementarities that are subsumed under these two categories, namely, "technological complementarities" and "innovational complementarities," was not proposed. The extended innovation ecosystems of digital platforms are already complex due to the heterogeneity and multiplexity[22] of the interested parties and stakeholders involved in them and such complexity will increase in the future. A better understanding of this taxonomy will thus be essential for digital

[22] That is, the different number and type of roles a stakeholder as an agent can play in these extended ecosystems.

platforms to generate innovation strategies allowing them to benefit from a dramatic increase not only in technological but also in innovational complementarities.

Modularization Modularization is also addressed in the third revision of the Teecean model and some of the pros and cons of modularization are discussed. This discussion is important for digital platform because it can shed light on the trade-off between incremental innovation based on modularization under an existing architecture and radical innovation based on a new architectural design. Contrary to the description of the platform Google Play as having been driven through modularization with a focus on incremental innovation and imitation (Teece 2018a, p. 1382), this strategy shows how a modular design with open APIs and clearly defined interfaces allowed the stakeholders in the extended innovation ecosystem around Google Play to catch up with the Apple Store in record time. This strategy was not driven by incremental innovation and imitation. Quite the opposite in fact. It was driven by a radical business model innovation that allowed Google to ignite a process of value creation that resulted in the rapid adoption of Android and its Google Play platform as de facto standards in the mobile devices and services industry.

The business model chosen by Google for its Android operating system and its Google Play platform focused on value creation, was designed to catch up with the iOS mobile operating system and the Apple Store platform, and constituted quite a departure from the business model adopted by Apple with its focus on value capture. The business model innovation executed by Google corresponded to an architectural innovation that provided the necessary incentives for stakeholders in its innovation ecosystem to innovate and ignite high value creation by allowing them to take a significant share of profits. This case provides an example of how to resolve the trade-off between innovation based on modularization and innovation based on a new architectural design. While modularization was needed to achieve the goal of catching up with the functionality of Apple's iOS, Google's goal was not to imitate Apple's business model but radically change it through a new architectural design in order for its overall catch-up strategy to be successful.

While incremental and radical innovation via modularization and architectural design can peacefully coexist and complement each other, architectural design will generally have a much higher impact on igniting substantial and radical innovation and will also be subject to higher strategic uncertainty. Seen from the perspective of the global stakeholder capitalism model of digital platforms set out in Section 4, these innovation strategies can, and generally will, be driven not only by incumbent digital platforms but also by the stakeholders in their ecosystems, thereby allowing them to deal with radical uncertainty (Kay and King 2020).

Standards Regarding the final discussion on standards in the third revision of the Teecean model, digital platforms may de facto act as both standard development and standard setting organizations. But we do not see the need for them "to foreclose business models other than non-exclusive licensing" (Teece 2018a, p. 1383). In fact, licensing will not generally be the preferred revenue model chosen by digital platforms when dealing with the problem of setting and developing standards and driving industrial policy in their extended innovation ecosystems.

On the contrary, there is a priori no limitations as to what business models can be implemented and what revenue models can de deployed for value capture. Revenue models can in fact vary considerably. They can be based on transactions or fees for access to tools or services, as the case of Twitter discussed in Section 3 exemplifies. But they can also be based on indirect models involving interested parties and stakeholders other than those setting, developing and using standards, as the online ad syndication model pioneered by Google exemplifies.

The global stakeholder capitalism model of digital platforms has also important implications for strategy. We elaborate on some of these implications below.

5.3 Implications for strategy

Digital platforms depart from the notion of value creation as taking place in linear value chains to embrace the notion of value creation through open innovation in value creation systems (VCSs).[23] Although the plea for such a radical departure is not new (Normann 2001), this paradigm shift is yet to take place in the mainstream strategic management literature (Eisenhardt and Martin 2000; Teece 2007; Porter 2008) and is more often mentioned in connection with strategic management approaches that construe strategy as practice (Golsorkhi et al. 2015; Ramírez and Wilkinson 2016). Yet the ongoing platform revolution is changing this situation.

Digital platforms are now in the process of transitioning from value creation in open innovation systems (Chesbrough 2003) to value creation in what we have termed the extended innovation ecosystems of digital platforms. If digital platforms are able to sustain the model described in this article and continue to extend their innovation ecosystems, then more complex strategic management frameworks that construe strategy as a dynamic coevolutionary decision-making process involving multiple stakeholders, multiple objectives, and multiple criteria will become ubiquitous. The new strategy of digital platforms will also contribute to the consolidation of strategic management theories, frameworks and decision-aiding tools rooted in more complex strategy-as-practice approaches (Golsorkhi et al. 2015).

5.3.1 From competing to coopeting strategies

Under the global stakeholder capitalism model of digital platforms set out in Section 4, digital platforms deploy an array of "coopeting" strategies aimed at igniting innovation in their extended innovation ecosystems, often transferring the lion's share of profits, and sometimes even all the profits, to interested parties in these extended innovation ecosystems. This constitutes a radical departure from securing control over complementary and cospecialized assets to capture Schumpeterian rents and secure the lion's share of profits under weak appropriability regimes when

[23] Some digital platform scholars make the distinction between pipeline businesses and platform businesses to refer to these two notions of how value is created (Parker et al. 2016; van Alstyne et al. 2006).

the innovator "is disadvantageously positioned vis-à-vis independent owners of complementary assets" and "is excellently positioned versus imitators with respect to commissioning complementary assets" (Teece 1986, p. 297).

Digital platforms adopt these new coopeting strategies by positioning big data as the generalized complementary asset that is required in their extended innovation ecosystems to profit from innovation and by providing it to interested parties, enabling them as highly innovative firms and drivers of innovation. Digital platforms will often give away the Schumpeterian rents they may otherwise be able to capture in order to build and sustain such networked ecosystems.[24] In the new global stakeholder capitalism model of digital platforms, this departure from neo-Schumpeterian wisdom is often a matter of necessity, as the catch-up strategy of Google described in Section 3.3 exemplifies.

Due to the higher complexity of the extended innovation ecosystems of digital platforms and the dilemma of ecosystem extension, the execution of coopeting strategies will require that digital platforms develop and deploy new integrative frameworks for the governance of their extended innovation ecosystems. These extended governance frameworks will be needed to keep not only the risk of latent competitors at bay, including the risk of being subject to platform envelopment attacks by interested parties and stakeholders (Eisenmann et al. 2007). They will also be needed to respond to threats and attacks from a wide variety of stakeholders that will populate their extended innovation ecosystems, most notably civil society, which may increasingly pose a threat to the global stakeholder capitalism model of digital platforms going forward.

The extended governance of digital platforms will need to master the hard balancing act between genuinely treating all stakeholders "as end in themselves" and treating them "purely instrumentally in the pursuit of narrow interests" (Singer 2013, p. 316).

5.4 The future of digital platforms

The extended innovation ecosystems of digital platforms go beyond innovation networks, as we construe them in neo-Schumpeterian frameworks. They are best understood as political subsystems loosely embedded in the political and legal systems of nation-states due to the lack of tight digital legislation and regulation. As incumbent digital platforms continue to extend their innovation ecosystems, they will not only invade and disrupt the space of incumbents in the real economy but also play an increasingly important role in politics and international relations.

5.4.1 Regulating digital platforms

Models of business ethics have been studied in the past based on the political assumptions that firms make regarding the exploitation of market limitations, such as "monopolistic tendencies, lack of ability to pay, unpriced externalities,

[24] The catch-up strategy executed by Google to position Android as the most widely used mobile operating system brilliantly exemplifies such coopeting strategies.

information asymmetries, failure to provide public goods, and lack of distributive justice," on the one hand, and the delivery of human goods, such as "wealth, health, justice, and aesthetics," on the other (Singer 2013, p. 309). Digital platforms are not immune to criticisms regarding the exploitation of market limitations and they have been under attack mainly due to their monopolistic tendencies.

Digital platforms are special in that they can expand globally due in part to lack of regulation. They also seem to be partly immune to current antitrust legislation. Such immunity is the result of the current focus of antitrust legislation on issues of price with the aim of protecting end consumers. As their business models draw upon near-zero marginal costs, digital platforms are usually able to offer services to end consumers at highly competitive prices. Digital platforms also deliver high value to end consumers in the form of free services based on revenue models not based on charging end consumers but rather charging for their attention, such as the revenue models of Google or Facebook. For these reasons digital platforms are not a usual suspect of current antitrust legislation.

The exploitation of market limitations, including monopolistic tendencies, by dominant digital platforms comes, however, in a myriad of different and novel forms. As a result, recent scholarly work has proposed to amend current antitrust legislation to capture the ways in which dominant digital platforms can exploit market limitations. Scholars have even proposed to regulate them as public utilities to mitigate the exploitation of some of these limitations (Kahn 2018).

As shown in Fig. 1, dominant digital platforms tend to exploit these market limitations and compensate for them either proactively through the delivery of human and also public goods or retroactively through the negotiation of tax regimes with governments at different levels instead of refraining from exploiting these limitations or delaying their exploitation.[25] Although the threat of new antitrust legislation is here to stay, digital platforms are in a privileged position to keep this threat at bay through the execution of preemptive strategies such as the generation of public goods in the form of social innovations as a form of compensation or the provision of bespoke services to governments at various levels.

5.4.2 Keeping the status quo

Incumbent digital platforms act as regulators and policymakers to address market failures that arise in the innovation ecosystems they create and orchestrate. To this effect, they deploy today their own governance models as a crucial component of their strategy (Gawer and Cusumano 2002; Cusumano et al. 2019; Yoffie et al. 2019). Although efforts to more tightly embed digital platforms in the political and legal systems of the nation-states in which they operate have been increasing in the past, digital platforms have been able to keep the current loose digital legislation and regulation by deploying self-regulation and preemptive strategies, thereby retaining the control of their micro and meso economies.

[25] A case in point here are the negotiations of bespoke tax regimes of Airbnb with municipalities and city councils in large metropolitan areas (Benner 2016; Nieuwland and van Malik 2020).

Whether digital platforms will continue to succeed in upholding the negative corporate freedoms they enjoy today, and whether such freedoms will increase or decrease going forward, will depend on whether they succeed in implementing integrative frameworks for the extended governance of digital platforms at the intersection of strategy, business ethics, politics and international relations.

5.4.3 The extended governance models of digital platforms

The extension of the innovation ecosystems of digital platforms will make the command and control they once exerted through their own governance models fade away. This loss of control will increase the pressure for tighter digital legislation and regulation and will call for the deployment of a new form of governance in order to keep this threat at bay. Key components of this new form of governance, which we have termed the "extended governance of digital platforms," will be the deployment of a new array of preemptive and self-regulation strategies and the cooperation with key stakeholders, especially governments and regulatory bodies.

The future extended governance models of digital platforms will implement mechanisms of group decision and negotiation for social choice encompassing governments and regulators at supranational, regional and national level.[26] These mechanisms will not only be driven by the threat of tighter legal regulatory regimes on the part of governments and the demands of civil society organizations and digital ad-hocracies, which will be in part ignited, driven and influenced by the digital platforms themselves. More importantly, they will be driven by the future mediating and moderating role of digital platforms in geopolitics and geostrategy.

6 Conclusions

Although already omnipresent in today's economy, the model under which digital platforms operate today had not yet been postulated in the literature. Borrowing constructs at the intersection of corporate strategy, business ethics and politics, we have set out "the global stakeholder capitalism model of digital platforms." We have described the macro environmental trends that gave rise to this model, explained its main political assumptions, and shown how this new economic model departs from previous models that emerged as a result of globalization in the 1990s, such as the global stakeholder capitalism model and the global model of hypercompetition.

Based on the analyses of several cases of success and failure in driving the strategy of digital platforms, we have shown that the positioning of big data as a generalized complementary in the "extended innovation ecosystems" of digital platforms

[26] This is shown in Fig. 1 by the relationships of digital platforms with governments at various levels, including cities, provinces, nation-states, regions across nation-states, and governance bodies at supranational level (Lessig 2006). A prototypical case of this form of negotiation are the agreements between Airbnb and cities in Europe, in which bespoke taxation preempting the installation of norms banning Airbnb's short rentals from large metropolitan areas was negotiated to preserve the business model of Airbnb (Nieuwland and van Malik 2020).

emerges as a result of the need to deal with a new problem, namely, the conversion of increasingly large volumes of big data into information and ultimately into knowledge. As this problem is ridden with deep strategic uncertainty and lack of organizational alignment, incumbent digital platforms have been executing new "grand strategies" to deal with it. Key to understanding these grand strategies is the positioning of big data as a generalized complementary asset in the extended innovation ecosystems of digital platforms.

The strategy of digital platforms in these new extended innovation ecosystems is quite a departure from conventional strategic management frameworks rooted in the resource-based view of the firm (Barney 1991). The emergence of the extended innovation ecosystems of digital platforms and the strategic positioning of big data as a generalized complementary asset in them is novel and presents new challenges that are neither addressed by mainstream management frameworks (Penrose 1995; Porter 2008) nor by the dynamic capabilities framework (Teece 2014).

We have shown why the strategy of digital platforms and the way they profit from innovation departs from innovation frameworks rooted in neo-Schumpeterian economics (Teece 1986). We have described why the rents captured by digital platforms differ from the Schumpeterian rents described in these frameworks in rather fundamental ways. We have also addressed the problem of value capture, showing why the way in which digital platforms profit from innovation departs from the standard model of how firms profit from innovation (Teece 2006). Our analysis shows that a focus on value capture over value creation in the extended innovation ecosystems of digital platforms often proves problematic and suggests that a carefully orchestrated balance between value creation in the early phases of the innovation lifecycle and value capture in later phases better captures the way in which digital platforms drive innovation and profit from it. We have also shown why recent revisions to this standard model fail to capture the way in which digital platforms drive and execute strategy and profit from innovation in their extended innovation ecosystems (Teece 2018a, b).

Our contribution is timely and relevant given the grand strategies of innovation ecosystem extension that leading digital platforms are currently executing in order to enter and disrupt existing industries. Despite the efforts of incumbents in these industries to neutralize the threat posed by digital platforms by transforming their business models via digital transformation, the disruption ignited by digital platforms in their extended innovation ecosystems is making the global stakeholder capitalism model of digital platforms gain relevance at a pace that is unprecedented in history. In view of these developments, the global stakeholder capitalism model of digital platforms is not only here to stay but is also poised to dominate in the global context in the near future.

Whether the global stakeholder capitalism model of digital platforms can be sustained in spite of increasing pressures from several quarters to install tighter regulation and legislation will ultimately depend on the ability of digital platforms to adopt models of extended governance able to preempt digital regulation and legislation. This will change the objective of corporate strategy from the current pursuit of either sustainable competitive advantage (Porter 2008; Teece 2007) or temporary competitive advantage (D'Aveni 1994, 2010; D'Aveni et al. 2010) to the fulfillment of corporate multi-fiduciary duties at the intersection of strategy, business ethics, politics and international relations through the deployment of extended governance frameworks. As they will need to allow digital platforms to account and respond to heterogeneous groups of stakeholders in highly complex extended innovation ecosystems on an ongoing basis, these extended governance frameworks will play a crucial role in the strategy of digital platforms in the future.

Future work will focus on the development and implementation of extended governance frameworks allowing digital platforms to drive transformations toward higher degrees of sustainability in their extended innovation ecosystems. Such frameworks will be of the essence for the strategy of digital platforms going forward, as they will allow them to master the hard balancing act between pursuing market opportunities and exploiting known limitations of market systems, such as monopolistic tendencies, on the one hand, and delivering human, public and common goods in their extended innovation ecosystems and society at large, on the other. We will also work on the development of a strategic management framework based on the concept of metadynamic capabilities that goes beyond the current competing views of dynamic capabilities (Eisenhardt and Martin 2000; Teece 2007, 2014). The aim of this metadynamic capabilities framework will be to better capture the way digital platforms inform, drive and execute business strategy in their extended innovation ecosystems.

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Declarations

Conflicts of interest The authors declare that they have no conflict of interest.

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